

YUYANG DENG

matthewgo2009@gmail.com

EDUCATION AND ACADEMIC APPOINTMENT

Columbia University

Postdoctoral Research Scientist

Sep. 2024 - Dec. 2025

Hosted by Prof. Samory Kpotufe. Working on theory of transfer and multi-task learning.

The Pennsylvania State University, University Park

Aug. 2020 - Dec. 2024

Ph.D. in Computer Science and Engineering

Advisor: Prof. Mehrdad Mahdavi

Concentration: Federated Learning, Optimization and Learning Theory.

The Pennsylvania State University, University Park

Aug. 2018 - May 2020

M.S. in Computer Science and Engineering (transfer to PhD program)

Soochow University, Suzhou

Sep. 2014 - Jun. 2018

B.E. in Communication Engineering.

PROFESSIONAL SUMMARY

Machine Learning researcher with expertise in **model personalization**, **federated learning**, **transfer learning**, **large-scale optimization** and **LLM post-training**. Experienced in developing scalable algorithms and deploying ML systems for privacy-preserving, distributed, and multi-task learning in real-world settings. **Strong publication record** (10+ papers at NeurIPS/ICML/AISTATS/ICLR, 1200+ citations). [Google Scholar](#)

TECHNICAL SKILLS

Programming Python, C++, MATLAB, Linux, Bash, Git

ML Frameworks PyTorch, TensorFlow, JAX, Scikit-learn

Specialties Model Personalization, Multi-domain (on-device) Learning, Transfer Learning, Optimization, Robust and Privacy-preserving ML

WORK EXPERIENCE

Accenture, Accenture Center for Advanced AI

Advanced AI Research Scientist Manager

Jan. 2026 - Present

- Work on the self-evolving LLM agent training. We proposed a challenger model free optimization algorithm where we derive the closed form optimal challenger model policy, and use Metropolis-Hasting algorithm to simulate sampling from this distribution to generate questions. Then, we use these questions to train solver model to improve its reasoning capability.

Cruise

AI Research Intern - Long-tailed Machine Learning

May. 2023 - Aug. 2023

- Work on the detection and medication of long-tailed learning problem. We use the gradient inner product as a signal to detect the long-tailed samples, and we found that it can effectively capture the rare image samples. Then we propose a re-weighted mini-batch SGD which improves the model accuracy by 3 – 5% on the longtail CIFAR-10/100/ImageNet dataset, compared to baseline SGD.

Sony AI

AI Research Intern - Privacy-Preserving Machine Learning

May. 2022 - Aug. 2022

- Work on studying the generalization bound of adversarially robust domain adaptation via the lens of adversarial Rademacher complexity, which bridges the longstanding gap between practice and theory in adversarial learning and multi-domain learning.

SELECTED PUBLICATIONS

1. Adaptive Personalized Federated Learning

Yuyang Deng, Mohammad Mahdi Kamani, Mehrdad Mahdavi.
arXiv preprint 2020

Proposed one of the first *adaptive personalized FL (APFL)* algorithms, where each client jointly trains the mixing of the local and global models. Achieved 5–10% higher test accuracy than baseline and state-of-the-art personalized FL methods, with improved communication efficiency. [Paper](#), [Code](#)

2. Distributionally Robust Federated Averaging

Yuyang Deng, Mohammad Mahdi Kamani, Mehrdad Mahdavi.
In Conference on Neural Information Processing Systems (NeurIPS) 2020

Introduced *Distributionally Robust Federated Averaging (DRFA)*, a communication-efficient method to improve robustness in FL under heterogeneous data. Utilized a novel snapshotting scheme for efficient gradient history approximation. Improved robust accuracy by 10% over FedAvg and achieved $3\times$ speedup compared to other robust FL algorithms. [Paper](#), [Code](#)

3. Distributed Personalized Empirical Risk Minimization

Yuyang Deng, Mohammad Mahdi Kamani, Pouria Mahdavinia, Mehrdad Mahdavi.
In Conference on Neural Information Processing Systems (NeurIPS) 2023

Proposed a new *data- & system-aware* paradigm (PERM) for learning from heterogeneous data sources by designing personalized objectives weighted via distribution discrepancy estimation. Introduced a model shuffling technique for parallel optimization and adaptive complexity across devices. Improved test accuracy by 15% on synthetic data and 5–10% on real-world datasets compared to prior state-of-the-art. [Paper](#)

4. On the Generalization Ability of Unsupervised pre-training

Yuyang Deng, Junyuan Hong, Jiayu Zhou, Mehrdad Mahdavi.
In International Conference on Artificial Intelligence and Statistics (AISTATS) 2024

Analyzed generalization in unsupervised pre-training + fine-tuning via representation transferability and Rademacher complexity. Proposed a novel pre-training-stage regularizer, improving fine-tuning accuracy by $\sim 2\%$ over non-regularized and L2-regularized baselines on Masked AutoEncoder pre-training with few-shot fine-tuning. [Paper](#)

5. Mixed-Sample SGD: an End-to-end Analysis of Supervised Transfer Learning

Yuyang Deng, Samory Kpotufe.
In Conference on Neural Information Processing Systems (NeurIPS) 2025

Proposed *Mixed-Sample SGD*, a fully adaptive algorithm for supervised transfer learning. The method dynamically samples from source and target data during training without requiring prior domain statistics, while remaining computationally efficient. Achieved 5% higher test accuracy than single-domain training on CIFAR Dog vs. Cat and matched state-of-the-art performance with $5\text{--}10\times$ faster runtime compared to prior methods. [Paper](#)

AWARDS

Best Student Paper, KDD2023 FL4Data-mining workshop.

Best AI PhD dissertation at Penn State University.

ACADEMIC SERVICES

Reviewers for NeurIPS, ICLR, ICML, AISTATS, COLT, JMLR, TSP, TWC, TNNLS

INVITED TALK

Collaborative Learning from Multiple Sources

at CMU Optimization, Probability and Learning (OPAL) Lab.

Virtual

Feb. 2024

Distributed Personalized Empirical Risk Minimization

at 3rd Federated Learning Online Discussion.

Virtual

Oct. 2023

Distributed Personalized Empirical Risk Minimization

at KDD FL4Data-mining workshop.

Long Beach, CA

Aug. 2023

Adaptive Personalized Federated Learning

at Bravo Software Inc.

Virtual

June. 2020